

RICHIE RYULIE SUGANDA

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EDUCATION

University of Houston, Cullen College of Engineering

Houston, TX

Ph.D. in Electrical and Computer Engineering

Jan. 2024 – Expected Jan. 2028

NAIL Lab — Networked Autonomous & Intelligent Systems Laboratory

Advisor: Dr. Bin Hu

Research: Safe autonomous systems, multi-robot systems, safety-critical control, perception-aware planning, and uncertainty quantification

Universitas Katolik Parahyangan (UNPAR)

Bandung, Indonesia

B.S. in Mechatronics, Robotics, and Automation Engineering

2019 – Sep. 2023

GPA: 3.70/4.00

Thesis: Designed and built a CoreXY motion-control platform incorporating LQG control, MATLAB/Simulink validation, custom RP2040 electronics, and real-time hardware implementation.

PUBLICATIONS

1. **R. R. Suganda** and B. Hu, “Adaptive Risk-Certified Event-Triggered Replanning for Dynamic Navigation,” *Conference on Robot Learning (CoRL)*, 2026. **Acceptance Rate: 33%.**
2. F. Zhang, **R. Suganda**, J. Chen, W. Liu, H. Fu, B. Hu, and Q. Lin, “Neural-ESO: A Dual-Pathway Architecture for Provably Robust Learning-Based Control,” *IEEE Robotics and Automation Letters (RA-L)*, 2026.
3. **R. R. Suganda** and B. Hu, “Formation-Aware Adaptive Conformalized Perception for Safe Leader-Follower Multi-Robot Systems,” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026. **Acceptance Rate: 36%.**
4. **R. R. Suganda**, T. Tran, M. Pan, L. Fan, Q. Lin, and B. Hu, “Distributed Perception Aware Safe Leader Follower System via Control Barrier Methods,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 7794–7800. **Acceptance Rate: 38.67%.**
5. Y. Zhang, E. Tran, **R. Suganda**, B. Hu, L. Sun, and J. Chen, “Usability Evaluation of QGroundControl and a Novel Mission Planner Add-on for Multi-UAV Operations,” *AIAA SciTech Forum*, 2027. **Accepted — Student Paper Competition.**
6. T. Bautista, F. Wang, A. Govindarajan, L. Sun, Y. Zhang, J. Chen, **R. R. Suganda**, B. Hu, M. S. Poltowicz, and E. T. Chancey, “A High-Fidelity 3D Simulation Environment for Multi-Aircraft Operations,” *AIAA SCITECH Forum*, 2026.
7. **R. R. Suganda** and T. A. Tamba, “Modeling and Output Feedback Control Design of a CoreXY System,” *IEEE FORTEI International Conference on Electrical Engineering*, 2024, pp. 166–171.

Preprints and Late-Breaking Research

- T. Tran, **R. R. Suganda**, and B. Hu, “XiYOLO: Energy-Aware Object Detection via Iterative Architecture Search and Scaling,” *arXiv preprint*, 2026.
- **R. R. Suganda**, I. R. Santjoko, M. Pan, and B. Hu, “Distributed 3D Leader-Follower Formation Control with Field-of-View Safety via Control Barrier Functions,” *IEEE/RSJ IROS Late-Breaking Poster Session*, 2026.
- **R. R. Suganda** and B. Hu, “Toward Embedded LLM-Guided Navigation and Object Detection for Aerial Robots,” *IEEE ICRA Late-Breaking Session*, 2025.

RESEARCH EXPERIENCE

University of Houston — NAIL Lab

Research Assistant

Houston, TX

Jan. 2024 – Present

- Conduct research in **safe autonomous systems, multi-robot coordination, uncertainty-aware navigation, robust learning-based control, and human–multi-robot interaction.**
- Develop safety-critical planning and control methods using **Control Barrier Functions, Model Predictive Control, CVaR, conformal prediction, and event-triggered replanning.**
- Developed a risk-certified navigation framework achieving **98% collision-free operation and 86% navigation success** under distribution shift, with **9/10 successful hardware-in-the-loop trials.**
- Design and validate algorithms across simulation, hardware-in-the-loop experiments, and physical robotic platforms including **ROSbot 2 Pro mobile robots and aerial robotic systems.**
- Develop robotics and machine-learning systems using **Python, C++, ROS/ROS 2, PyTorch, numerical optimization, and GPU computing.**

SELECTED PROJECTS

Embedded AI and Edge Deployment

Jetson / MAX78xxx

- Develop and benchmark efficient machine-learning pipelines on **NVIDIA Jetson Orin and MAX78000/78002** embedded AI platforms.
- Work includes **model quantization, pruning, neural architecture search, inference profiling, and accuracy–energy tradeoff evaluation.**

ROSbot 2 Pro Embedded Compute Upgrade

Jetson Orin Nano, ROS2, Linux

- Upgraded the ROSbot 2 Pro computing platform from an UP Board to an **NVIDIA Jetson Orin Nano**, integrating the new embedded computer with the robot’s sensing, communication, and control stack.
- Ported and validated the ROS software stack to ARM64 Linux, integrating robot drivers, communication interfaces, sensing, control, and autonomous-navigation components on the Jetson Orin Nano.

TEACHING EXPERIENCE

University of Houston

Teaching Assistant — ELET 4397

Houston, TX

Fall 2026

- Support instruction in **embedded AI and efficient deep learning**, including PyTorch model development, object detection, pruning, quantization, neural architecture search, and edge deployment.
- Develop laboratory material involving **MAX78000/78002, NVIDIA Jetson, YOLO, TinyML, inference profiling, and accuracy–efficiency benchmarking.**

Universitas Katolik Parahyangan

Bandung, Indonesia

Teaching Assistant — Automatic Control Systems

Sep. 2023 – Dec. 2023

Student Laboratory Assistant — Automatic Control Systems

Feb. 2023 – Jul. 2023

Teaching Assistant — Engineering Mathematics

Feb. 2022 – Sep. 2022

TECHNICAL SKILLS

Programming: Python, C++, MATLAB, Bash, L^AT_EX, CUDA

Robotics & Control: ROS/ROS 2, Control Barrier Functions, Model Predictive Control, multi-robot systems, formation control, autonomous navigation, path planning, trajectory replanning, SLAM

Machine Learning: PyTorch, object detection, conformal prediction, uncertainty quantification, neural architecture search, evolutionary optimization, LoRA, DoRA, pruning, quantization, OpenCV

Robotic & Embedded Platforms: ROSbot 2 Pro, ModalAI Seeker UAV, ModalAI Sentinel UAV, CrazyFlie 2.1 Brushless, NVIDIA Jetson Orin, MAX78000/78002, Raspberry Pi RP2040

Hardware & Prototyping: PCB design, KiCad, soldering, micro-soldering, electronics prototyping, hardware debugging, embedded systems integration

Mechanical Design & Fabrication: SolidWorks, 3D CAD, mechanical design, FDM and resin 3D printing, fiber laser operation, diode laser cutting/engraving, rapid prototyping

Tools & Methods: Linux, Git, Bash, OPNsense, Docker, llama.cpp, LMStudio, Hermes

AI Tools: ChatGPT, Codex, Gemini, Claude Code, Cursor

RESEARCH INTERESTS

Safe Autonomous Systems · Safety-Critical Control · Multi-Robot Systems · Autonomous Navigation · Human–Robot Interaction · Perception Uncertainty · Efficient Machine Learning · Embedded AI